decomposing vegetation in wetlands and termite effluences. Meanwhile, in places where free oxygen is scarce, methane does not always require life to form. At this very moment, astrobiologists are arguing over the exact origin of trace methane on Mars and the copious quantities of methane on Saturn's moon Titan, where cows and termites we presume do not dwell.

If the aliens track our nighttime side while we orbit our host star, they might notice a surge of sodium from the widespread use of sodium-vapor streetlights that switch on at dusk in urban and suburban municipalities. Most telling, however, would be all our free-floating oxygen, which constitutes a full fifth of our atmosphere.

Oxygen—which, after hydrogen and helium, is the third most abundant element in the cosmos—is chemically active and bonds readily with atoms of hydrogen, carbon, nitrogen, silicon, sulfur, iron, and so on. It even bonds with itself. Thus, for oxygen to exist in a steady state, something must be liberating it as fast as it's being consumed. Here on Earth, the liberation is traceable to life. Photosynthesis, carried out by plants and many bacteria, creates free oxygen in the oceans and in the atmosphere. Free oxygen, in turn, enables the existence of oxygen-metabolizing life, including us and practically every other creature in the animal kingdom.

We Earthlings already know the significance of our planet's distinctive chemical fingerprints. But distant aliens who come upon us will have to interpret their findings and test their assumptions. Must the periodic appearance of sodium be technogenic? Free oxygen is surely biogenic. How about methane? It, too, is chemically unstable, and yes, some of it is anthropogenic, but as we've seen, methane has nonliving agents as well.

If the aliens decide that Earth's chemical features are sure evidence of life, maybe they'll wonder if the life is intelligent. Presumably the aliens communicate with one another, and perhaps they'll presume that other intelligent life-forms communicate, too. Maybe that's when they'll decide to eavesdrop on Earth with their radio telescopes to see what part of the electromagnetic spectrum its inhabitants have mastered. So, whether the aliens explore with chemistry or with radio waves, they might come to the same conclusion: a planet where there's advanced technology must be populated with intelligent life-forms, who may occupy themselves discovering how the universe works and how to apply its laws for personal or public gain.